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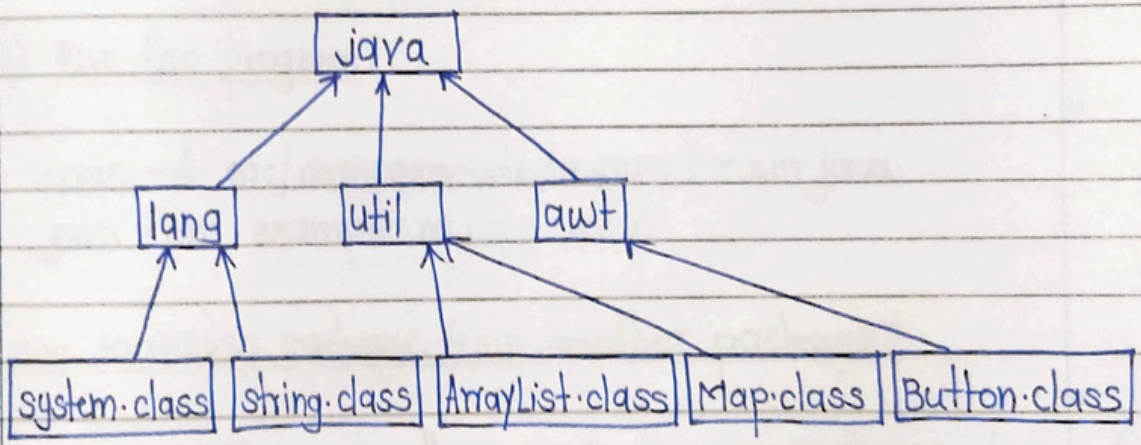
9 · Java Encapsulation

9.1 Java package:

A java package is a group of similar types of classes, interfaces, subpackages. Packages in java can be categorized in two form, built-in packages and user-defined packages. There are many built-in packages such as java, lang, awt, javax, swing, net, io, util, sql, etc.

Advantages of Java package:

- Java package is used to categorize the classes and interfaces so that they can be easily maintained.
- Java package provides access protection.
- Java package removes naming collision



Creation of Java packages:

1) Package name - 'com.example.myapp'.

2) Create directory structure:
mkdir -p src/com/example/myapp

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3) Write a class:

```
package com.example.myapp;
  public class Myclass{
    public void display(){
      System.out.println("Hello from myclass!");
  }
}
```

4) Compile the code:

```
javac -d .src/com/example/myapp/*.java
```

5) Run the program:

```
Java com.example.myapp.Main
```

6) Run the program:

```
Javac -d. src/com/example/myapp/Main.java
java com.example.myapp.main.
```

How to access package from another package?

There are three ways to access the package from outside the package.

1. import package.*;
2. import package.classname
3. fully qualified name.

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9.2 Access Modifiers in Java.

The access modifiers in java specifies the accessibility or scope of a field, method, constructor, or class. We can change the access level of fields, constructors, methods, and class by applying the access modifer on it.

There are four types of Java access modifiers:

1) private:

Private access modifier is accessible only within the class

Example:

```
class A{
 private int data=40;
 private void msg(){ system.out.println("Hello java");}
 }
 public class simple{
  public static void main(string args[]){
   A obj = new A();
   System.out.println(obj.data);
   obj.msg();
  }
 }
```

2) Default:

If you don't use any modifier, it is created as default by default. The default modifier is accessible

only within package.

3) Protected:

The protected Access modifier is accessible within package and outside the package but through inheritance only. The protected access modifier can be applied on the data member, method, constructor. It can't be applied on the class. It provides more accessibility than the default modifier.

4) Public:

The public access modifier is accessible everywhere. It has the widest scope among all other modifiers.

9.3 Encapsulation in Java:

Encapsulation in Java is a process of wrapping code and data together into a single unit, for example a capsule which is mixed of several medicines.

Advantages of Encapsulation

By providing only a setter or getter method, you can make the class read-only or write-only. In otherwords, you can skip the getter and setter methods.

It provides you the control over the data. Suppose you want to set the value of id which should be greater than 100 only, you can write the logic inside the setter method You can write the logic not to store the negative numbers in the setter methods.

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It is a way to achieve data hiding in Java beacause other class will not be able to access the data through the private data members.

The encapsulates class is easy to test, so it is better for unit testing.